YU JU (EDWIN) CHEN, PhD

Senior Software Engineer | CUDA, GPU Kernels & High-Performance Computing

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SUMMARY

GPU software engineer with a PhD in Computer Science and 5+ years of research and production experience building performance-critical C++/CUDA software. Experience includes CUDA kernel optimization, warp-level algorithms, coalesced memory layouts, shared-memory caching, kernel fusion, host-device synchronization reduction, GPU numerical methods, and real-time simulation.

CORE EXPERTISE

GPU Performance: CUDA; GPU kernel optimization; warp-level programming; coalesced memory access; shared memory; kernel fusion; memory-layout optimization; synchronization reduction; performance profiling
Profiling Tools: NVIDIA Nsight Systems; NVIDIA Nsight Compute
Parallel/Numerical: C++; DirectX/HLSL compute; NVIDIA Warp; sparse linear systems; LDL^T; CG; implicit/semi-implicit solvers; floating-point robustness; parallel algorithms
Production Software: Python; C#; Rust; Unreal Engine; Buck; CMake; Linux; Git; P/Invoke/FFI; API/ABI compatibility; CI/testing

EXPERIENCE

Software Engineer, Digital Humans | Meta Reality Labs – *Aug 2024–Present*

- Optimized GPU deformable-simulation kernels through vectorization, removal of redundant GPU–CPU synchronization, and device-buffer preallocation; tuned adaptive timesteps and regularization for production workloads.
- Integrated NVIDIA Warp/Newton releases across API/ABI changes for cloth, tetrahedral soft bodies, and XPBD constraints, resolving production compatibility and numerical-correctness issues.
- Parallelized Unreal Engine mesh loading and vertex-buffer transfer and implemented triple buffering between simulation and render threads, reducing game-thread mesh-transfer time from 12 ms to 5 ms (58%).
- Built and maintain production C++ numerical code and parity tests for bit-level cross-language equivalence; hardened epsilon-sensitive interpolation, quaternion edge cases, NaNs, and non-normalized inputs.

Senior Researcher, Graphics & Vision | Tencent America – *Feb 2022–Aug 2024*

- Implemented and optimized CUDA sparse-system construction and LDL^T solution for thousands of independent banded systems using specialized GPU layouts, warp-level operations, shared-memory caching, coalesced access, and kernel fusion.
- Achieved up to 10× speedup over a naive GPU implementation for 3,000 systems and up to 186× over Eigen CPU LDL; integrated the solver into a real-time GPU hair-simulation pipeline in C++/CUDA.
- Developed the real-time hair-simulation plug-in for Unreal Engine 5 using CUDA and DirectX compute shaders; pipeline reached up to 260 FPS for 2,000+ simulated strands on RTX 3080 and was published at SCA 2023 with Best Paper Honorable Mention.

Research Software Engineer / Founding Engineer | Rapidia Tech Inc. – *Jul 2019–Jan 2022*

- Built 3D visualization and printer-control software from the ground up; owned architecture spanning numerical geometry, device control, and customer workflows.

Research Intern | Adobe Creative Technologies Lab – *May 2017–Aug 2017*

- Developed a PCA-based reduced model for consistent soft-body dynamics across mesh resolutions; work published at SCA 2019.

SELECTED NUMERICAL RESEARCH

- PhD research in stiff elastodynamics and implicit/exponential integration; routinely solved sparse systems with Intel MKL PARDISO and Eigen, generally below 100k unknowns.
- Implemented Conjugate Gradient and worked with Lanczos and Arnoldi methods for iterative and eigenvalue computations.

EDUCATION

PhD, Computer Science | University of British Columbia – *2014–2020*
Advisors: Uri Ascher and Dinesh Pai. Dissertation: *Integrators for elastodynamic simulation with stiffness and stiffening.*

BASc, Engineering Physics | University of British Columbia – *2009–2014*

SELECTED PUBLICATIONS

Towards Realtime: A Hybrid Physics-Based Method for Hair Animation on GPU – *SCA 2023 – Best Paper Honorable Mention*
SIERE: A Hybrid Semi-Implicit Exponential Integrator for Efficiently Simulating Stiff Deformable Objects – *ACM TOG 2020*
EigenFit for Consistent Elastodynamics Simulation Across Mesh Resolution – *SCA 2019*
Exponential Rosenbrock-Euler Integrators for Elastodynamic Simulation – *IEEE TVCG 2017*